

Craig Barber

Software Engineer | Cleveland, OH | seawyvern@gmail.com |
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Seasoned software engineer with backgrounds in full software development life-cycle, cloud infrastructure, technical leadership and developer community outreach. Passionate about engineering excellence, providing a first class customer experience, and empowering developers.

Professional and Technical Skills

- Served as technical lead for SWE team providing mentorship, technical direction, and code/design reviews.
- Developed software with Java, C#, Python, Javascript, Typescript, Go, Bash, HTML, and CSS.
- Experience working on cloud platforms including Google Cloud Platform (GCP) and Microsoft Azure.
- Experience using Linux, MacOS, and Windows.
- Developed DevOps solutions using Terraform, Github Actions, Jenkins, Docker, Kubernetes, and Helm.
- Experience working with MySQL, MSSQL, NoSQL, and Cloud Storage systems.

Work Experience

Google - Software Engineer 2016-2022

Google Cloud Platform, Open Source Infrastructure, SWE/TL

Led design and implementation of integrations between GCP and popular open source infrastructure tools including Terraform, Jenkins, Github Actions, and Cloud Foundry.

- Led development and support for GCP's official Jenkins integrations:
 - GCE: <https://github.com/jenkinsci/google-compute-engine-plugin>
 - GCS: <https://github.com/jenkinsci/google-storage-plugin>
 - OAuth: <https://github.com/jenkinsci/google-oauth-plugin>
 - GCR: <https://github.com/GoogleCloudPlatform/jenkins-gcr-plugin/>
 - GKE: <https://github.com/jenkinsci/google-kubernetes-engine-plugin>
 - Jenkins is the most widely used system for software build and test and accounts for the most CI/CD usage on GCE.
- Served as technical lead for Cloud Graphite's CI/CD GCP integrations team.
 - Worked with external partners building strong company ties.

- Engaged with CloudBees helping to secure their participation in the GCP Anthos Partner program.
- Met with customers gathering requirements.
- Created development roadmaps.
- Developed HTTP load balancer support for GCP Terraform provider:
<https://github.com/hashicorp/terraform-provider-google>
- Designed, implemented, and published Jenkins GKE Plugin.
- Designed, deployed, and operated internal team's Jenkins CI/CD infrastructure.
- Engaged in external developer community building and engagement:
 - DevOps World US 2019 tech talk, Jenkins on GKE:
https://www.youtube.com/watch?v=I8B65trrrEA&list=PLvBBnHmZuNQKLGcpoYh8pCfEez43_l8Ts&index=34&t=0s
 - DevOps World EU 2019 tech talk, Jenkins X Optimized for GCP:
<https://devopsworldjenkinsworld2019lisbo.sched.com/event/VFHp/jenkins-x-optimized-for-gcp>
 - DevOps World EU 2019 workshop, Hands-on with Cloud Native CI-CD:
<https://devopsworldjenkinsworld2019lisbo.sched.com/event/XTlc/hands-on-with-cloud-native-cicd>
 - Served on governance committee for Jenkins X:
<https://github.com/jenkins-x/steering>
 - Published Jenkins X GCP Playbook:
<https://github.com/GoogleCloudPlatform/jenkins-integration-samples/tree/master/jenkins-x>
- Designed, implemented, and published setup-gcloud plugin for Github actions v2.0:
<https://github.com/google-github-actions/setup-gcloud>
 - This plugin is actively used by over 170,000 customer projects on GCP, accounting for over 4 billion API requests over each 90 day period.
- Bootstrapped the GCP GitHub Actions 20% project, which now includes contributions from engineers across GCP teams.

Kythe, SWE

Kythe is a team which develops and maintains a scalable graph-structured semantic index of source code. Providing a graph querying service covering all of Google's internal source code.

- Designed and implemented Bazel's Build language indexer for Kythe: <https://kythe.io/>
 - The data generated by this indexer serves as the backbone for Bazel's Build language support in Google's internal code search and language semantic services.
- Designed and implemented ETL pipelines for importing Bazel's Build language semantic index data into the Kythe graph.
- Designed and implemented a prototype for Kythe Maven Extractor.

Microsoft - Software Development Engineer 2015-2016

Azure Security Forensics

Azure Security Forensics is a team which provides scalable/automated forensics analysis services for Azure VMs.

Security SDE II, SDE Intern (Summer 2014)

- Designed and implemented a system which orchestrated ETL pipelines for consuming and Indicators of Compromise (IOC) data from security partners, transforming said data into a structured IOC semantic database.
- Designed and implemented a system which automated the extraction of forensics data from Azure VHDs, cross-referencing extracted data against the IOC semantic database to generate alerts of potentially compromised customer VMs.
 - This service is used to continually scan over 220,000 active Azure VMs detecting malicious use and saving customers from fraudulent charges.

Apple - Software Engineer Intern Summer 2013

Hardware Engineering Reliability, SWE Intern

Apple's Hardware Engineering Reliability is a team which develops hardware reliability tests for all Apple hardware products, and maintains ETL pipelines consuming said test data for analysis.

- Designed and implemented tools which provided importation and analysis of internal hardware reliability/testing data.
 - These tools are utilized during the hardware testing process to verify manufacturing QA of iPhones, Macbooks, and other Apple products.

University of Arizona - Web Developer 2011-2014

- Designed and implemented web applications internally for labs within the Neuroscience Department, as well as publicly for the Center for Insect Science:
<https://cis.arl.arizona.edu/home>

Knowledge Computing Corporation - Software Engineer

2008-2011

- Designed and implemented ETL pipelines for law enforcement database services:
<https://www.redbooks.ibm.com/redpapers/pdfs/redp5353.pdf>

Tekco Management Group - Web Developer 2007-2008

- Designed and implemented web services for affiliate marketing and video streaming services.

Octopi Media Design Labs - Game Developer 2004-2007

- Designed and implemented mobile games which were published and pre-installed on Nokia's S40 and S60 devices.
- Served as TL leading design, implementation, and support of back-end service for the multiplayer web based game PoxNora.

Community Contributions

Girls In Engineering Math and Science (GEMS) - Google Coding Workshops

- Directed an annual coding on-site workshop program for GEMS at Google:
<https://seattleawis.org/gems/>

- Lead volunteer organization, logistics, and funding efforts.

GCP Adopt A School Hackathon Mentor

- Served as a mentor representing GCP at the University of Arizona's annual "Hack Arizona" hackathon.

Education

- B.S. Computer Science, University of Arizona, Tucson, AZ, 2014

References

- Evan Brown, Software Engineering Manager, Google Cloud GPU Networking, Google, evanbrown@google.com
- Joseph Lewis III, Senior Software Engineer, Google Cloud GPU Networking, Google, joseph@josephlewis.net
- Averil Kitsch, Staff Engineer, Database AI MCP Integrations, Google, akitsch@google.com
- Adam Powell, Senior Staff Engineer, Android Developer Experience, Google, adamp@google.com
- Daniel Moy, Site Reliability Engineer, File Solutions, Google, daniel.moyo30@gmail.com
- Dave Stanke, Specialist Customer Engineer, Application Modernization, Google, davidstanke@google.com
- Svetlana Gaivoronski, Senior Director of Security Engineering, SAS, Oracle, s.Gaivoronski@gmail.com